

EDA revision

2025-12-29

```
getwd() ## get the working directory
```

```
## [1] "C:/Users/Elaina.Flynn/OneDrive - GV-WFL Boces - Edutech/Documents"
```

```
## Load libraries
```

```
suppressMessages(library(tidyverse))
```

```
suppressMessages(library(forcats))
```

```
library(car)
```

```
## Loading required package: carData
```

```
##
```

```
## Attaching package: 'car'
```

```
## The following object is masked from 'package:dplyr':
```

```
##
```

```
##      recode
```

```
## The following object is masked from 'package:purrr':
```

```
##
```

```
##      some
```

```
library(summarytools)
```

```
##
```

```
## Attaching package: 'summarytools'
```

```
## The following object is masked from 'package:tibble':
```

```
##
```

```
##      view
```

```
## Load data
```

```
school_front <- read.csv("Job Applicants DATA 12-29.csv") ## recruitment info.
```

```
View(school_front)
```

```
## Demographic
```

```
GPA <- na.omit(school_front$Under.Graduate.Major.GPA)
```

```
Job.Title <- na.omit(school_front$Job.Title)
```

```
City <- school_front$Permanent.City
```

```
n <- 5816
```

```

job_title <- school_front$Job.Title

job_department <- school_front$Department

summary(job_title) ## the summary

```

```

##      Length      Class      Mode
##      5816 character character

```

These are descriptive statistics; it describes the distribution of groups

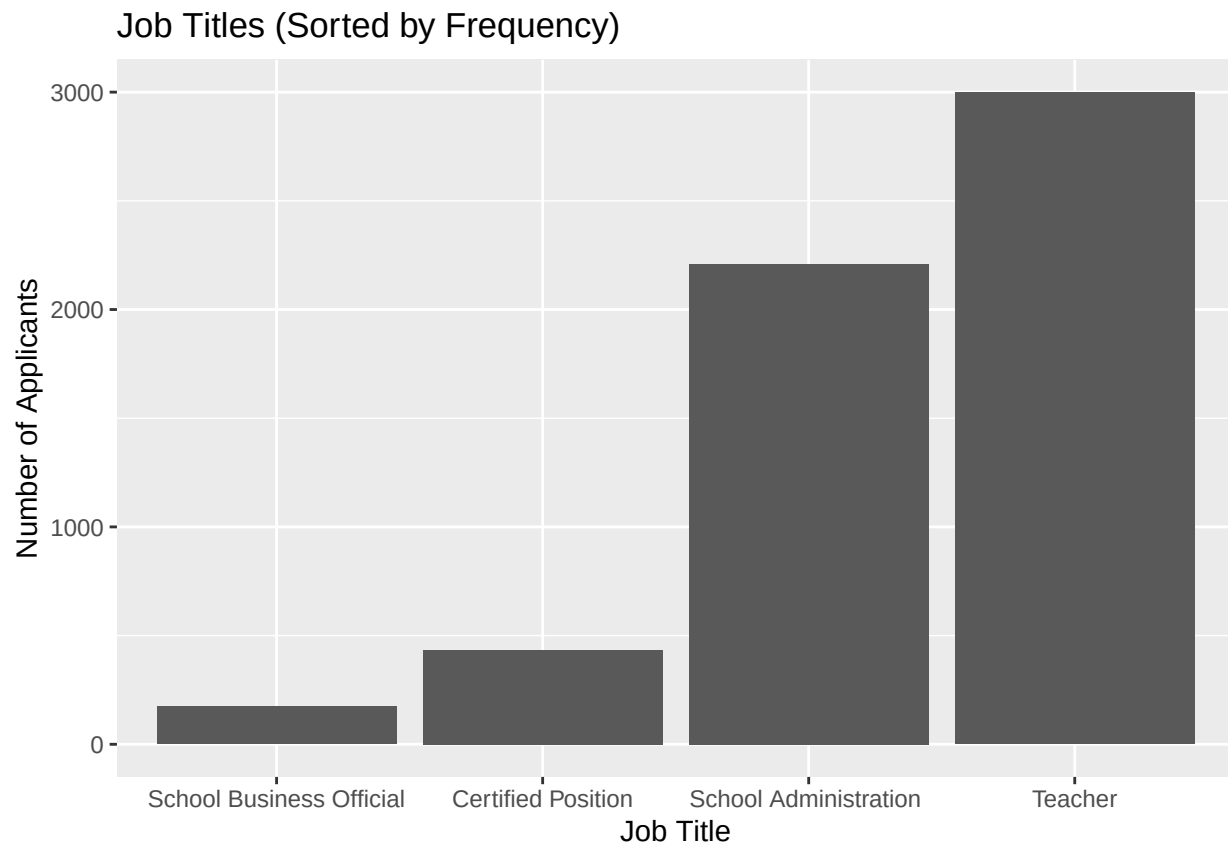
Job Titles Distribution

```

job_count <- school_front %>%
  count(Job.Title)

ggplot(job_count, aes(x = fct_reorder(Job.Title, n), y = n)) +
  geom_col() +
  labs(
    x = "Job Title",
    y = "Number of Applicants",
    title = "Job Titles (Sorted by Frequency)"
  )

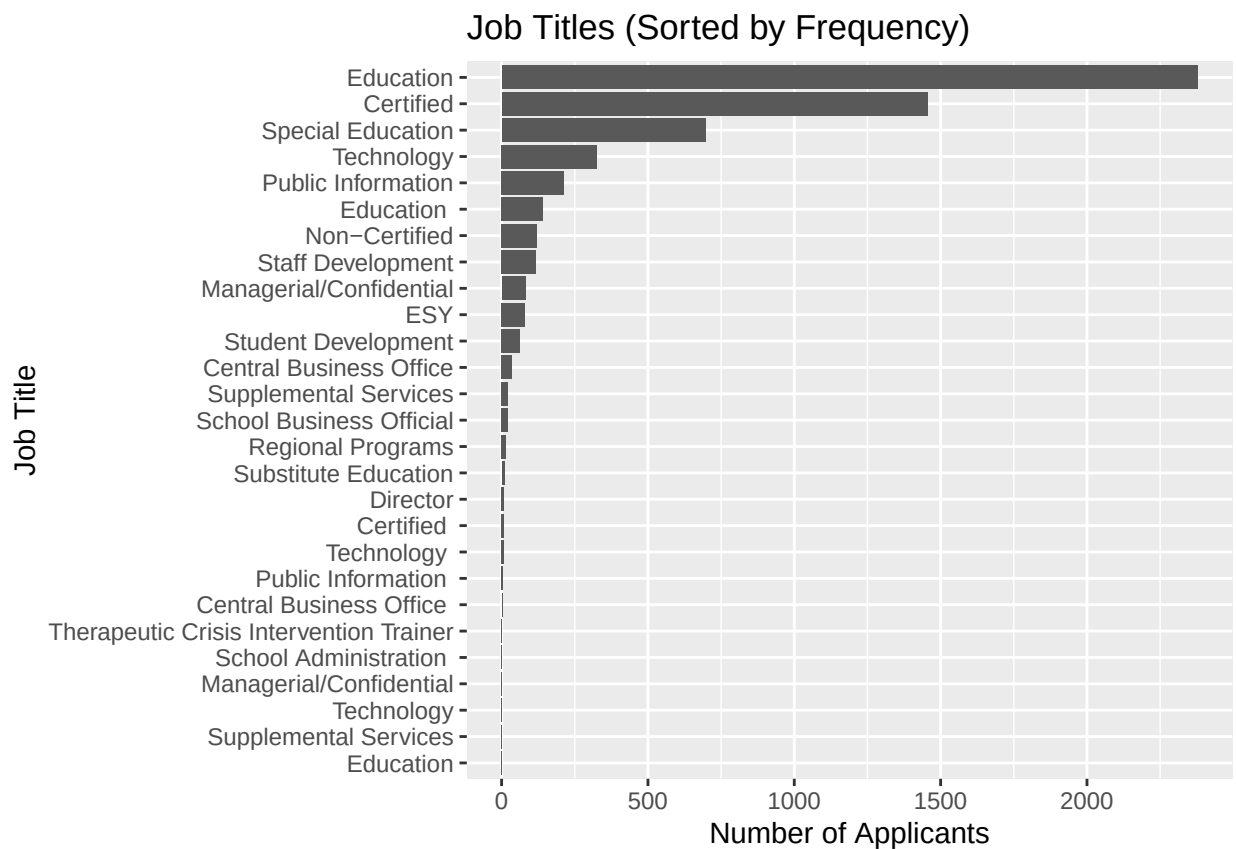
```



```
job_box_plots <- ggplot(data = school_front, mapping = aes(x = Job.Title)) +  
  geom_boxplot()
```

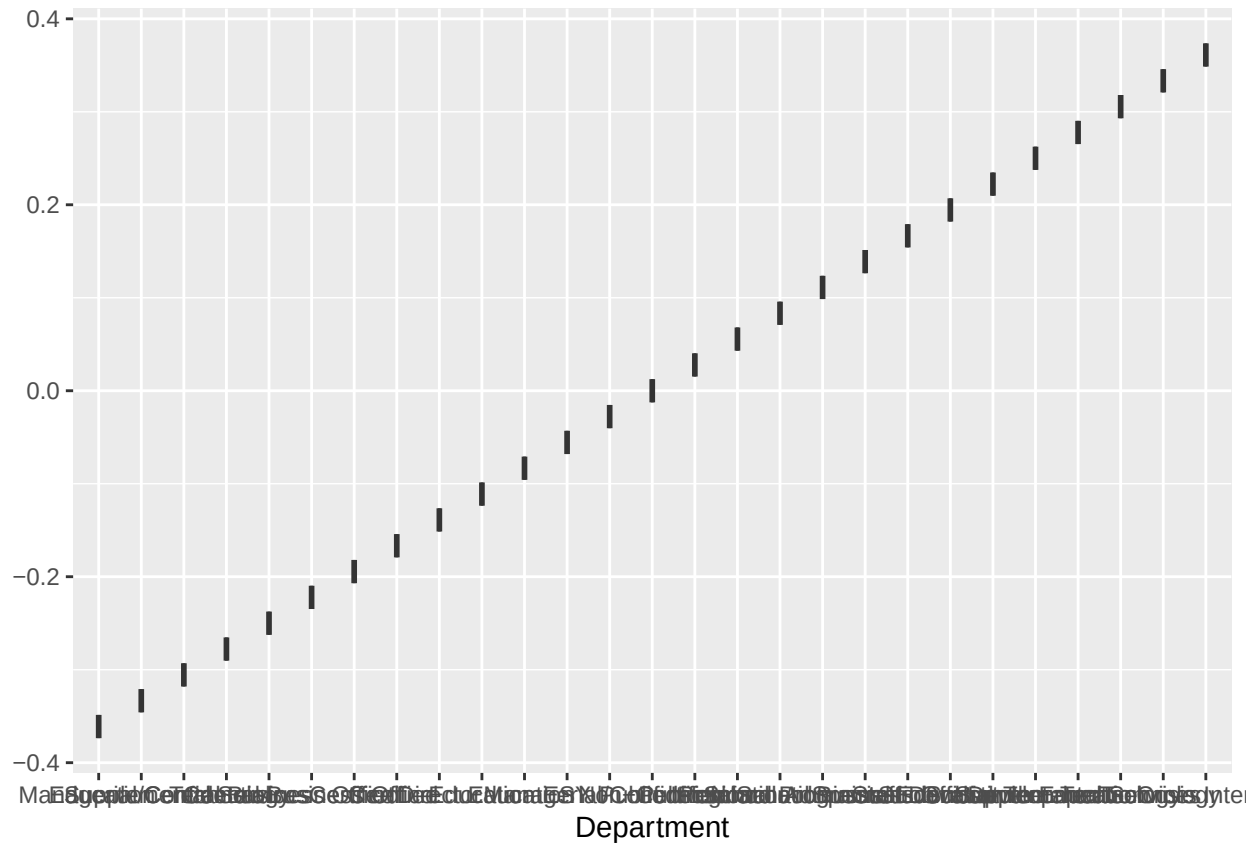
```
job_count <- school_front %>%  
  count(Department)
```

```
ggplot(job_count, aes(x = fct_reorder(Department, n), y = n)) +  
  coord_flip() +  
  geom_col() +  
  labs(  
    x = "Job Title",  
    y = "Number of Applicants",  
    title = "Job Titles (Sorted by Frequency)"  
  )
```



```
department_box_plots <- ggplot(data = school_front, mapping = aes(x = Department)) +  
  geom_boxplot()
```

```
department_box_plots
```



GPA Quantiles

```
## Quantiles tell you what to expect for distribution
```

```
quantile(GPA, 0.25) ## First Quartile
```

```
## 25%
```

```
## 3.2
```

```
quantile(GPA, 0.50) ## Second Quartile
```

```
## 50%
```

```
## 3.5
```

```
quantile(GPA, 0.75) ## Third Quartile
```

```
## 75%
```

```
## 3.8
```

```
IQR(GPA) ## 0.6 difference between Q1 (3.2) & Q3 (3.8). Meidan is 3.5
```

```
## [1] 0.6
```

```

standard_deviation <- sd(GPA)

variance <- var(GPA)

## Co-efficient of Variation
standard_deviation / mean(GPA)

```

```
## [1] 0.1159501
```

Numerics

```

Private.Years <- na.omit(school_front$Private.Years)

Public.Years.Teaching <- na.omit(school_front$Public.Years.Teaching)

Public.Years <- na.omit(school_front$Public.Years)

```

Cross-Tabulations

```

ctable(x = Public.Years.Teaching[1:n], y = Job.Title[1:n], chisq = TRUE) ## Display results of Chi-squa

```

```
## Error in table(names(candidates))["tested"]: subscript out of bounds
```

```

## Warning in parse_call(mc = match.call(), var = "x", var_label = FALSE, caller =
## "ctable"): metadata extraction terminated unexpectedly; inspect results
## carefully

```

```
## Error in table(names(candidates))["tested"]: subscript out of bounds
```

```

## Warning in parse_call(mc = match.call(), var = "y", var_label = FALSE, caller =
## "ctable"): metadata extraction terminated unexpectedly; inspect results
## carefully

```

```

## Warning in chisq.test(freq_table_min): Chi-squared approximation may be
## incorrect

```

```

## Cross-Tabulation, Row Proportions
## Public.Years.Teaching * Job.Title
##

```

```

## -----
##      Job.Title   Certified Position   School Administration   School Business O
## Public.Years.Teaching
##      0.00000000      104 ( 6.9%)      610 ( 40.3%)      53
##      0.25000000      0 ( 0.0%)      0 ( 0.0%)      0
##      0.30000000      0 ( 0.0%)      0 ( 0.0%)      0
##      0.50000000      0 ( 0.0%)      3 ( 37.5%)      0

```

##	1.00000000	18 (6.7%)	89 (33.2%)	7
##	1.25000000	0 (0.0%)	0 (0.0%)	0
##	1.50000000	1 (6.2%)	8 (50.0%)	0
##	2.00000000	27 (9.3%)	105 (36.3%)	5
##	2.50000000	0 (0.0%)	1 (16.7%)	0
##	2.75000000	0 (0.0%)	0 (0.0%)	0
##	3.00000000	18 (6.9%)	97 (37.5%)	11
##	3.50000000	1 (4.3%)	2 (8.7%)	1
##	4.00000000	8 (5.1%)	67 (42.4%)	7
##	4.50000000	0 (0.0%)	1 (25.0%)	0
##	4.75000000	1 (33.3%)	2 (66.7%)	0
##	4.80000000	0 (0.0%)	1 (100.0%)	0
##	5.00000000	12 (5.3%)	80 (35.2%)	7
##	5.50000000	0 (0.0%)	2 (28.6%)	0
##	5.80000000	1 (33.3%)	0 (0.0%)	0
##	6.00000000	21 (7.0%)	123 (41.1%)	8
##	6.50000000	0 (0.0%)	3 (42.9%)	0
##	6.60000000	0 (0.0%)	1 (100.0%)	0
##	7.00000000	7 (3.9%)	76 (42.0%)	4
##	7.50000000	0 (0.0%)	0 (0.0%)	0
##	7.60000000	0 (0.0%)	0 (0.0%)	0
##	8.00000000	12 (5.4%)	88 (39.8%)	2
##	8.50000000	0 (0.0%)	2 (50.0%)	0
##	9.00000000	14 (9.2%)	52 (34.2%)	4
##	9.40000000	0 (0.0%)	0 (0.0%)	0
##	9.50000000	2 (66.7%)	0 (0.0%)	0
##	9.70000000	0 (0.0%)	1 (100.0%)	0
##	10.00000000	22 (8.4%)	104 (39.5%)	5
##	10.50000000	0 (0.0%)	1 (16.7%)	0
##	10.60000038	0 (0.0%)	0 (0.0%)	0
##	11.00000000	5 (4.2%)	49 (41.2%)	7
##	12.00000000	23 (13.2%)	51 (29.3%)	4
##	12.50000000	0 (0.0%)	2 (28.6%)	1
##	13.00000000	6 (5.2%)	45 (38.8%)	5
##	13.50000000	0 (0.0%)	3 (75.0%)	0
##	14.00000000	15 (12.9%)	38 (32.8%)	5
##	14.50000000	2 (28.6%)	4 (57.1%)	0
##	15.00000000	16 (8.8%)	75 (41.4%)	11
##	15.50000000	1 (50.0%)	1 (50.0%)	0
##	16.00000000	10 (10.9%)	36 (39.1%)	0
##	16.50000000	0 (0.0%)	0 (0.0%)	0
##	17.00000000	10 (10.6%)	27 (28.7%)	1
##	17.30000019	0 (0.0%)	0 (0.0%)	0
##	18.00000000	7 (5.0%)	53 (37.6%)	2
##	18.50000000	0 (0.0%)	1 (100.0%)	0
##	19.00000000	9 (12.3%)	24 (32.9%)	2
##	19.50000000	0 (0.0%)	0 (0.0%)	0
##	20.00000000	13 (12.7%)	30 (29.4%)	4
##	20.80000000	0 (0.0%)	0 (0.0%)	0
##	21.00000000	4 (6.6%)	26 (42.6%)	2
##	21.50000000	0 (0.0%)	0 (0.0%)	0
##	22.00000000	4 (8.0%)	19 (38.0%)	3
##	22.50000000	0 (0.0%)	1 (25.0%)	0
##	23.00000000	5 (7.9%)	15 (23.8%)	2

```
##          23.50000000      0 ( 0.0%)      2 ( 50.0%)      0
##          24.00000000      2 ( 6.5%)      8 ( 25.8%)      1
##          24.60000000      0 ( 0.0%)      0 (  0.0%)      0
##          25.00000000      3 (10.0%)     12 ( 40.0%)      2
##          26.00000000      3 (10.7%)     12 ( 42.9%)      0
##          27.00000000      1 ( 2.5%)     25 ( 62.5%)      0
##          28.00000000      3 (23.1%)      2 ( 15.4%)      1
##          28.50000000      0 ( 0.0%)      0 (  0.0%)      0
##          29.00000000      0 ( 0.0%)      6 ( 42.9%)      0
##          30.00000000      2 ( 7.4%)     12 ( 44.4%)      1
##          30.50000000      0 ( 0.0%)      0 (  0.0%)      0
##          31.00000000      0 ( 0.0%)      4 (100.0%)      0
##          32.00000000      0 ( 0.0%)      2 ( 66.7%)      0
##          33.00000000      1 (14.3%)      1 ( 14.3%)      0
##          34.00000000      0 ( 0.0%)      8 ( 66.7%)      0
##          35.00000000      0 ( 0.0%)      1 (100.0%)      0
##          36.00000000      0 ( 0.0%)      2 ( 50.0%)      0
##          38.00000000      0 ( 0.0%)      2 (100.0%)      0
##          40.00000000      0 ( 0.0%)      4 ( 80.0%)      0
##          42.00000000      0 ( 0.0%)      0 (  0.0%)      0
##          43.00000000      0 ( 0.0%)      1 (100.0%)      0
##          <NA>          20 ( 8.4%)      86 ( 36.3%)      6
##          Total        434 ( 7.5%)     2209 ( 38.0%)     174
## -----
## -----
## Chi.squared   df    p.value
## -----
##    244.9654    234    0.2981
## -----
```

non-significant p value, these groups are independent (null hypothesis supported)

Public Teaching Experience (Box Plots)

```
## Q1 (lower 25%)
quantile(Public.Years.Teaching, 0.25) ## Lower 25% (0 years, new teachers)
```

```
## 25%
##    0
```

```
## Q2 (median)
quantile(Public.Years.Teaching, 0.50) ## Median 50% (6 years of experience)
```

```
## 50%
##    6
```

```
## Q3 (upper 75%)
quantile(Public.Years.Teaching, 0.75) ## Upper 75% (12 years)
```

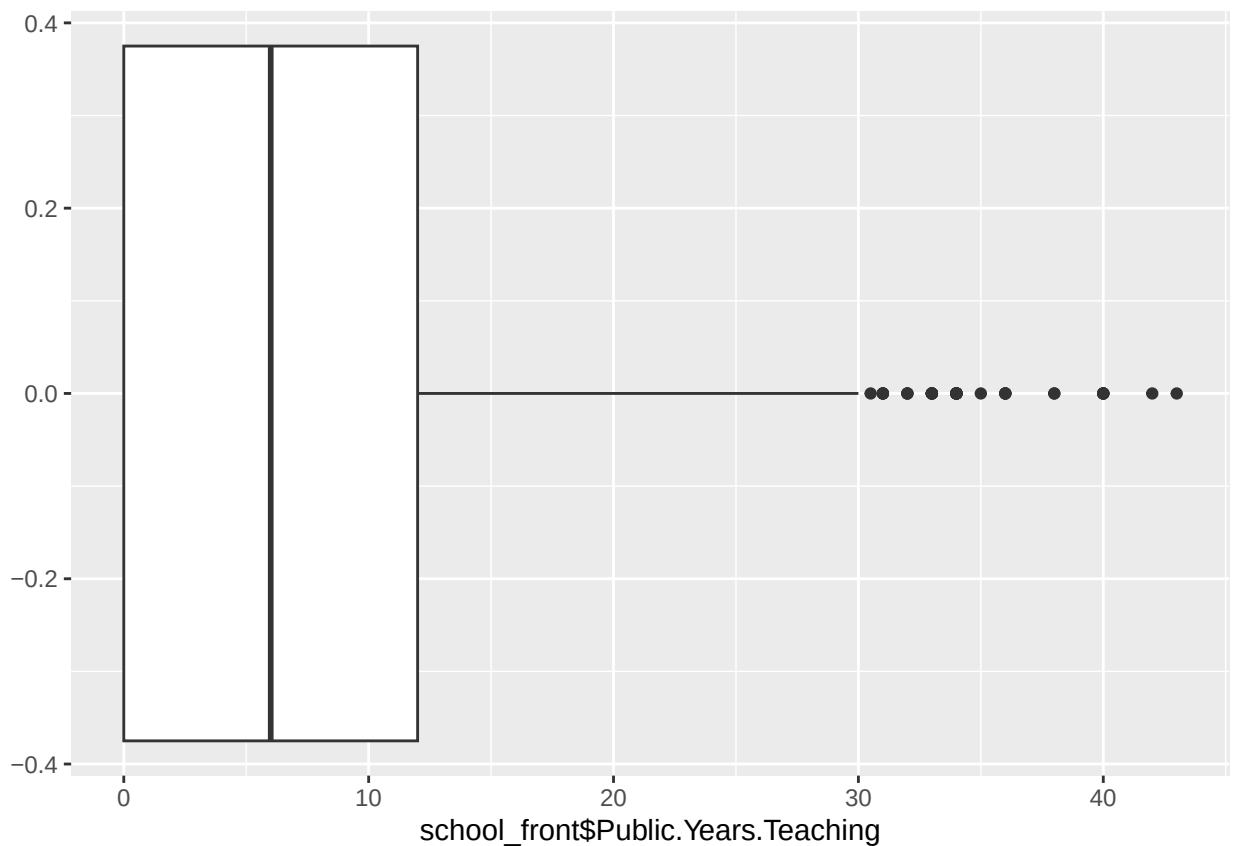
```
## 75%  
## 12
```

```
box_plots <- ggplot(data = school_front, mapping = aes(x = school_front$Public.Years.Teaching)) +  
  geom_boxplot()
```

```
box_plots
```

```
## Warning: Use of 'school_front$Public.Years.Teaching' is discouraged.  
## i Use 'Public.Years.Teaching' instead.
```

```
## Warning: Removed 237 rows containing non-finite outside the scale range  
## ('stat_boxplot()').
```



```
## Skewed right (means unusual values at higher end, not as common)
```

```
## Average teacher has 6 years experience
```

Experience General

```
## Q1 (lower 25%)  
quantile(Public.Years, 0.25) ## Lower 25% (0 years, new teachers)
```

```
## 25%
## 0
```

```
## Q2 (median)
quantile(Public.Years, 0.50) ## Median 50% (3 years of experience)
```

```
## 50%
## 3
```

```
## Q3 (upper 75%)
quantile(Public.Years, 0.75) ## Upper 75% (12 years)
```

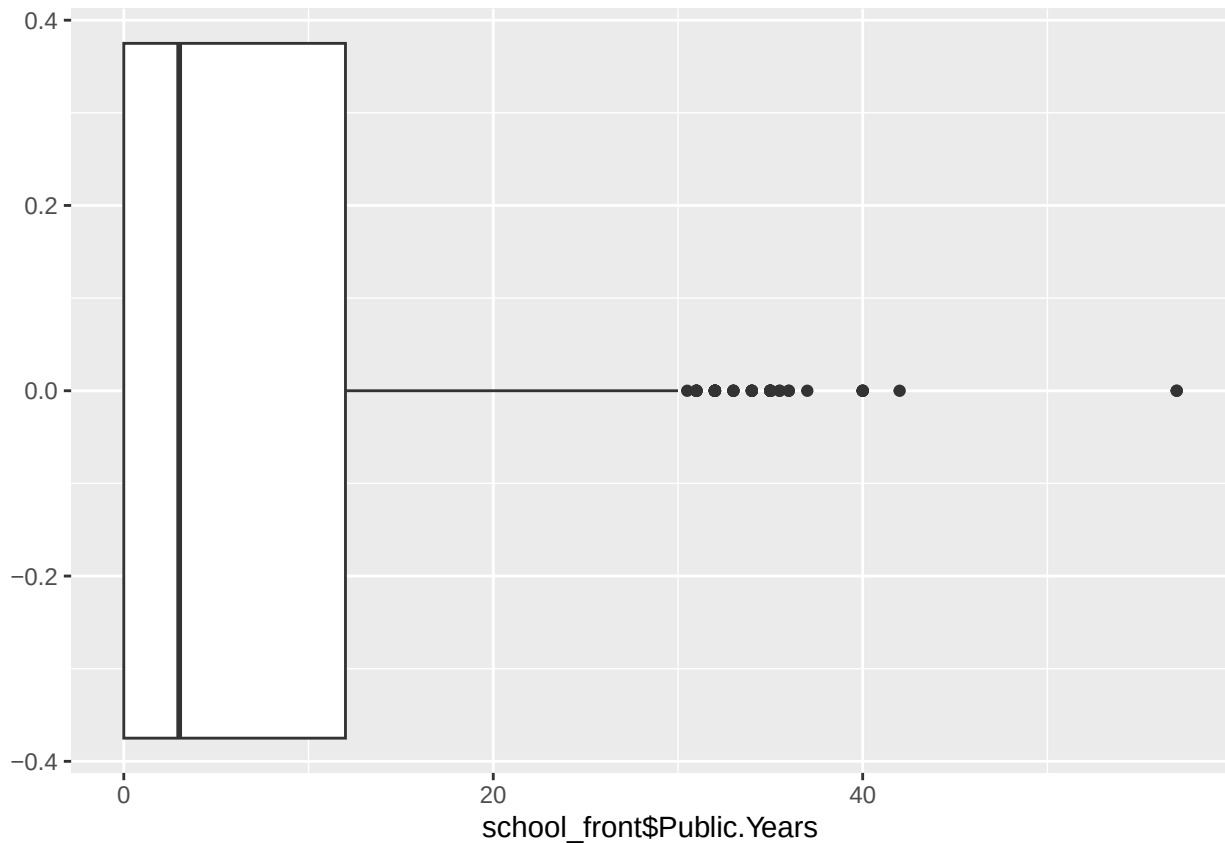
```
## 75%
## 12
```

```
box_plots <- ggplot(data = school_front, mapping = aes(x = school_front$Public.Years)) +
  geom_boxplot()
```

```
box_plots
```

```
## Warning: Use of 'school_front$Public.Years' is discouraged.
## i Use 'Public.Years' instead.
```

```
## Warning: Removed 260 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```



Correlation

```
n <- 5816

## Are years of experience and GPA correlated?
cor(Public.Years.Teaching[1:n], GPA[1:n], use = 'complete.obs') ## Not correlated (no relation)

## [1] -0.004334458

## Are private experience and GPA correlated?
cor(Private.Years[1:n], GPA[1:n], use = 'complete.obs') ## Not correlated (no relation)

## [1] -0.005406193

## Are public years and GPA correlated?
cor(Public.Years[1:n], GPA[1:n], use = 'complete.obs') ## No

## [1] -0.0004283961

## Are public and private years correlated?
cor(Public.Years[1:n], Private.Years[1:n], use = 'complete.obs') ## No

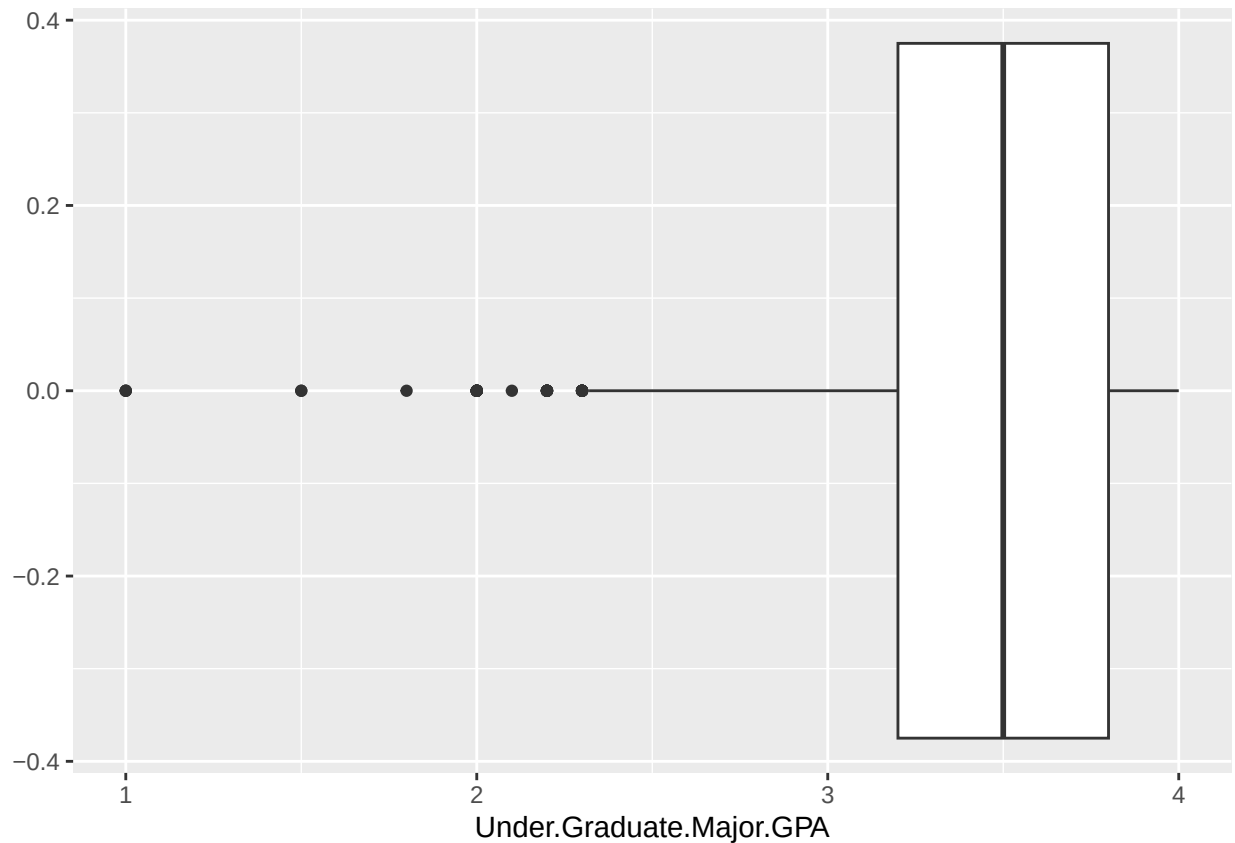
## [1] 0.002518986
```

GPA Boxplot Distribution

```
box_plots <- ggplot(data = school_front, mapping = aes(x = Under.Graduate.Major.GPA)) +
  geom_boxplot()

box_plots

## Warning: Removed 2360 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```



```
## Median is 3.5 gpa (Min. = 1.0; Max. = 4.0)

## Skewed left with outliers; the left whisker is longer than the right

## Distribution has left skew

## IQR is (3.25 - 3.75)

## Interpretation: Interpretation: A longer left tail suggests that a few smaller values are pulling the mean down

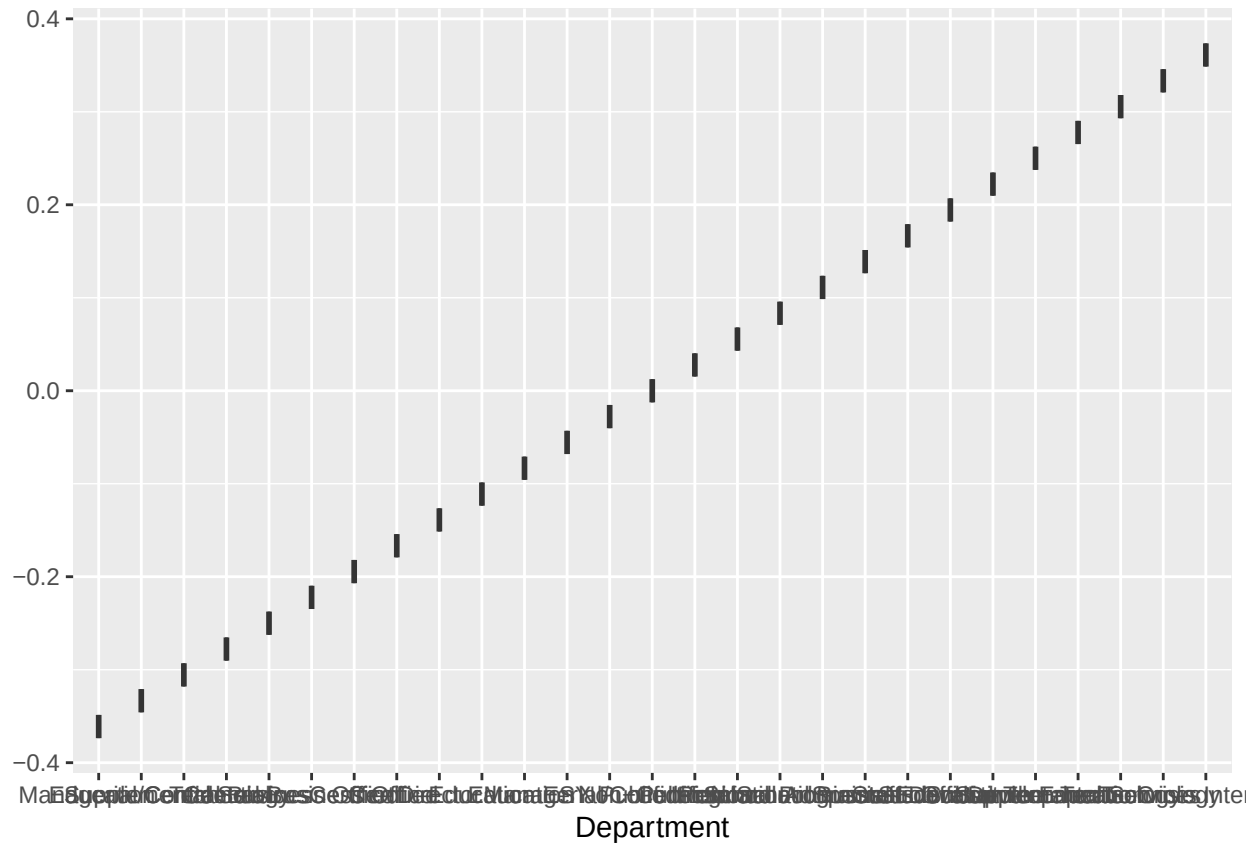
## Mean is less than median (the mean incorrectly represents data because of left skew; **use the median)

## This sample has an above average GPA at baseline... The average GPA is a 3.0
```

GPA as numeric?

```
box_plots <- ggplot(data = school_front, mapping = aes(x = Department)) +
  geom_boxplot()

box_plots
```

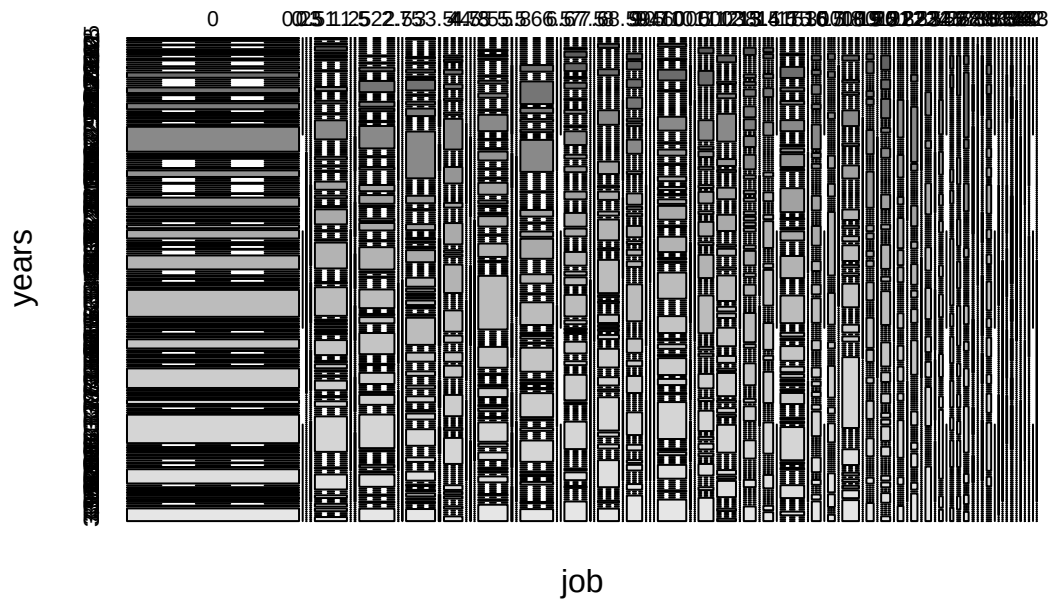


Mosaic Plots

```
## Try Job / Years Experience

mosaicplot(table(Public.Years.Teaching[1:n], GPA[1:n]),
  color = TRUE,
  xlab = "job", # label for x-axis
  ylab = "years" # label for y-axis
)
```

```
table(Public.Years.Teaching[1:n], GPA[1:n])
```



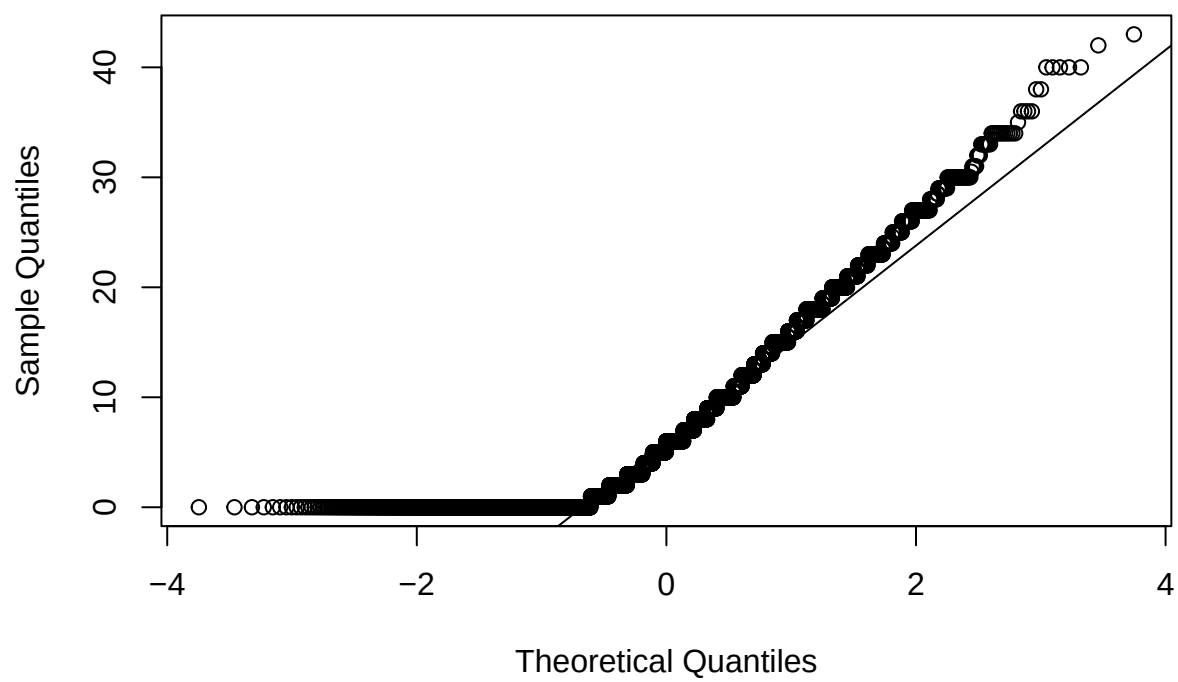
Test for Time-Series

```
## Job Title Time-Series
```

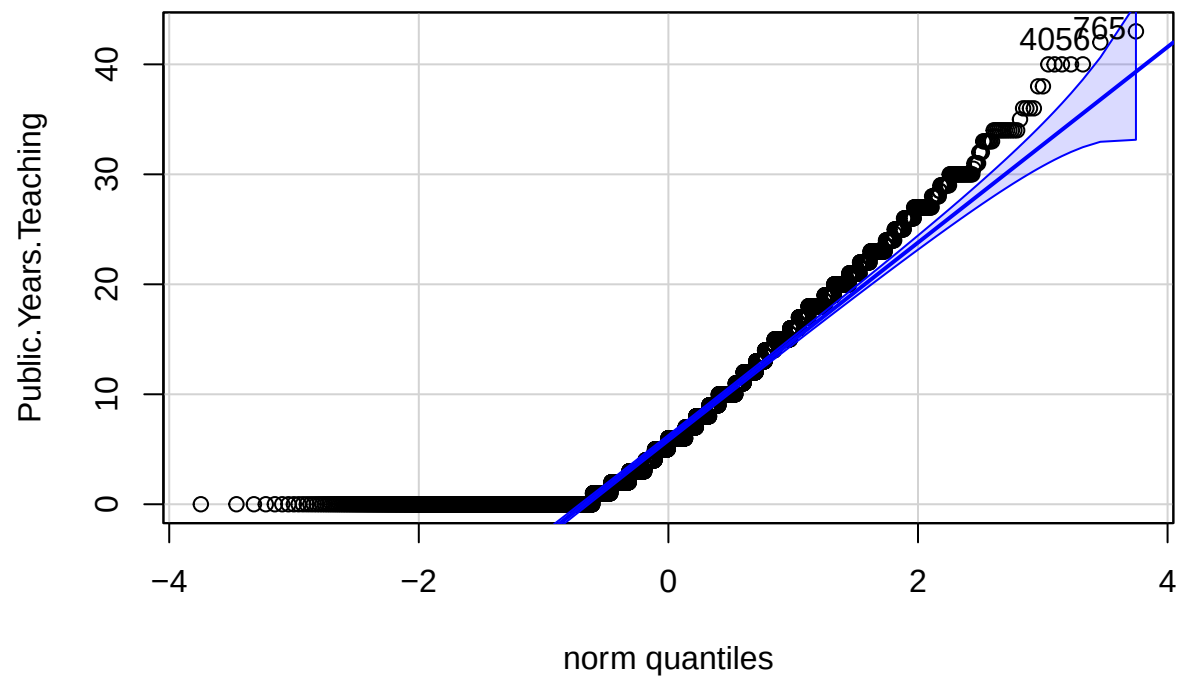
QQ Plots (check normality assumptions)

```
qqnorm(Public.Years.Teaching)
qqline(Public.Years.Teaching) ## Deviates at upper-tail
```

Normal Q-Q Plot

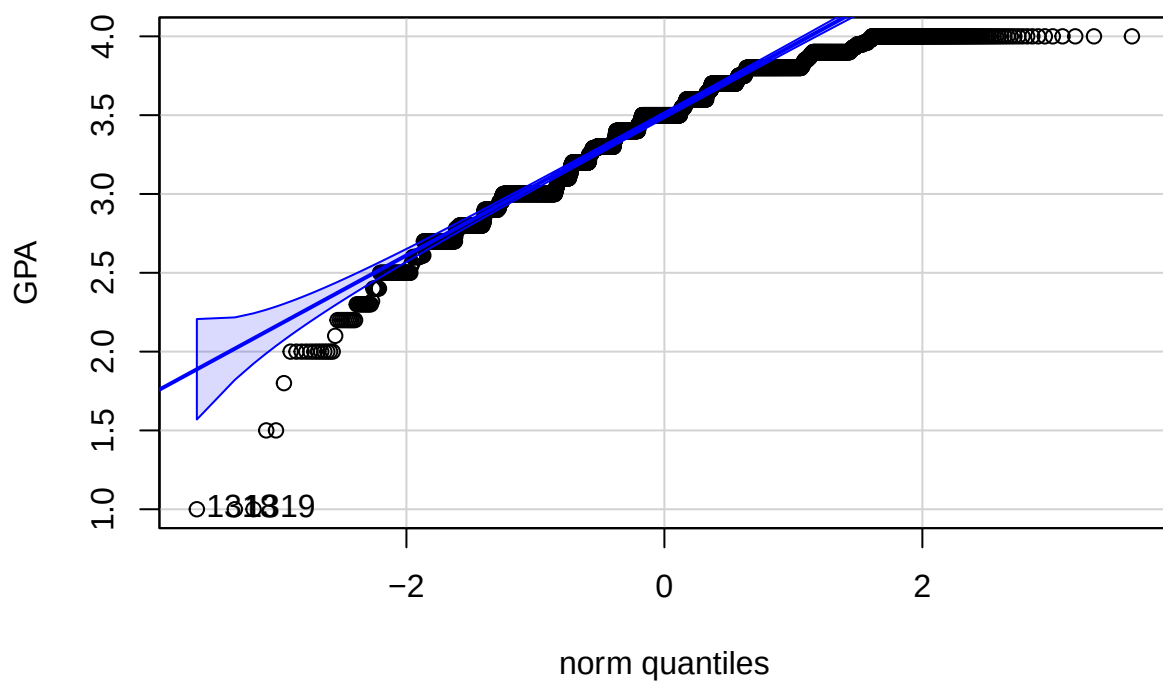


```
qqPlot(Public.Years.Teaching)
```



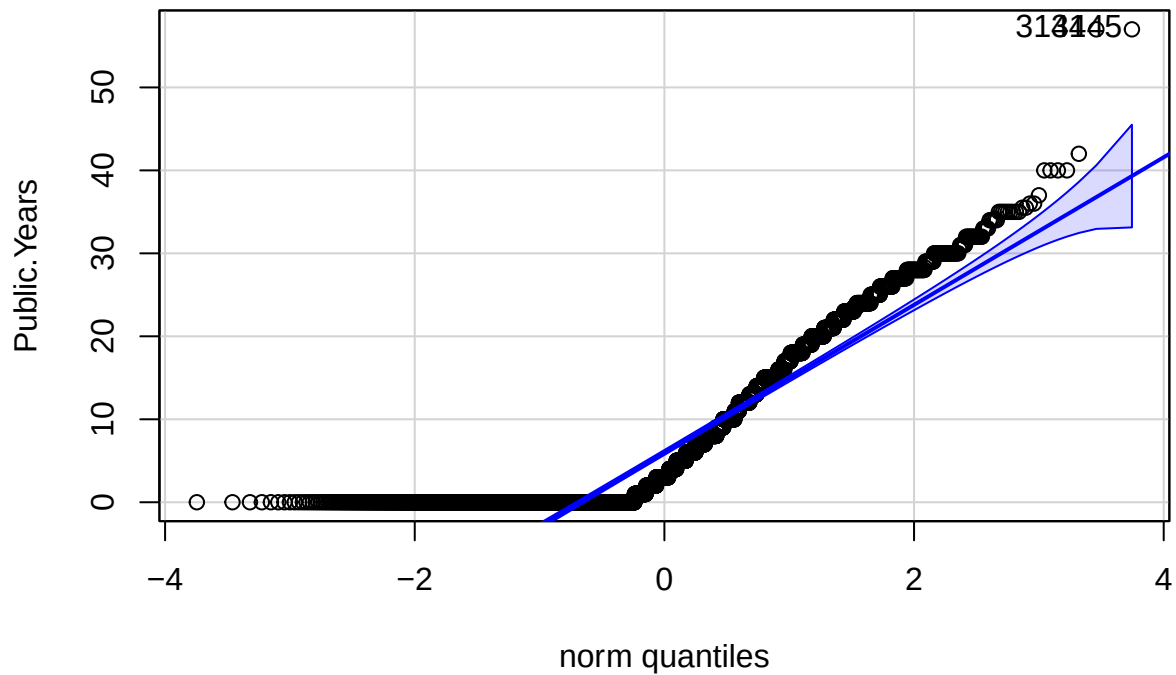
```
## [1] 765 4056
```

```
qqPlot(GPA)
```



```
## [1] 1318 1319
```

```
qqPlot(Public.Years)
```



```
## [1] 3144 3145
```

Correlation Matrix

Try t-test for job titles?

```
t.test(Public.Years.Teaching) ## Sig. difference based on years of experience
```

```
##
## One Sample t-test
##
## data: Public.Years.Teaching
## t = 71.784, df = 5578, p-value < 2.2e-16
## alternative hypothesis: true mean is not equal to 0
## 95 percent confidence interval:
## 7.287315 7.696517
## sample estimates:
## mean of x
## 7.491916
```

```
t.test(GPA) ## Sig. difference based on GPA
```

```
##
## One Sample t-test
##
## data: GPA
## t = 507.01, df = 3455, p-value < 2.2e-16
## alternative hypothesis: true mean is not equal to 0
## 95 percent confidence interval:
## 3.430544 3.457180
## sample estimates:
## mean of x
## 3.443862
```

```
t.test(Public.Years)
```

```
##
## One Sample t-test
##
## data: Public.Years
## t = 61.025, df = 5555, p-value < 2.2e-16
## alternative hypothesis: true mean is not equal to 0
## 95 percent confidence interval:
## 6.831397 7.284874
## sample estimates:
## mean of x
## 7.058136
```

```
t.test(Private.Years)
```

```
##
## One Sample t-test
##
## data: Private.Years
## t = 26.31, df = 5555, p-value < 2.2e-16
## alternative hypothesis: true mean is not equal to 0
## 95 percent confidence interval:
## 1.290522 1.498319
## sample estimates:
## mean of x
## 1.394421
```